Longitudinal Effects of Mantra-Based Meditation on Functional Connectivity

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1. INTRODUCTION

Background: Japa meditation involves vocal or silent mantra repetition to maintain focused attention.
Motivation: Previous research found that Hare Krishna (HK) meditation maintained higher individual alpha frequency (IAF) after training, whereas Sa-Ta-Na-Ma (SA) showed significant alpha slowing [1]. However, changes in alpha frequency alone do not characterize functional connectivity between brain regions.
Objective: Investigate changes in frontoparietal alpha-band (8–12 Hz) functional connectivity following six weeks of HK or SA meditation.

2. EXPERIMENTAL DESIGN [2]

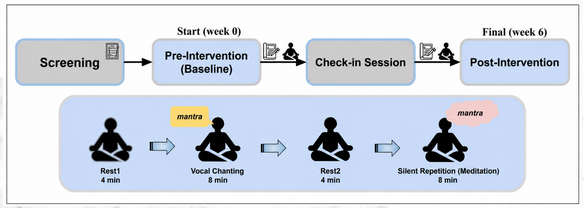


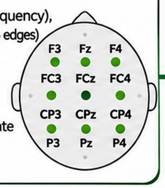
Figure 1. Experimental design and meditation-session protocol.

Participants: 31 participants with complete PRE and POST EEG data, assigned to HK (n = 16) or SA (n = 15).
Training: Six weeks of vocal and silent practice, increasing from 5 to 15 minutes per day; adherence tracked through online journals.
EEG Sessions: PRE (Week 0) and POST (Week 6), including eyes-closed rest, vocal chanting, and silent meditation.

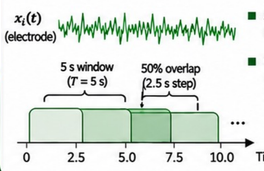
3. METHODS

a. EEG Data & Conditions

- **EEG:** 250 Hz (sampling frequency), 12 electrodes (unique pairs: 66 edges)
- **Conditions:** PRE / POST intervention, meditation
- **Segments:** Meditation minutes 2–6, full resting-state recordings



b. EEG Windowing

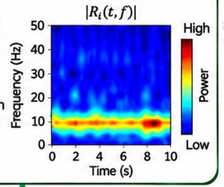


- **Alpha:** 8–10, 10–12 Hz (5-s windows, 95 windows)
- **Gamma:** 30–50 Hz (400-ms windows, 200-ms steps, 1,199 windows)

c. RID Time–Frequency Analysis [4]

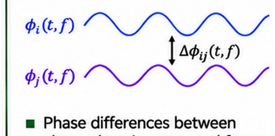
$$R_i(t,f) = A_i(t,f)e^{j\phi_i(t,f)}$$

- RID–Rihaczek transforms EEG into a complex time–frequency representation with instantaneous amplitude $A_i(t,f)$ and phase $\phi_i(t,f)$.



d. Pairwise Phase Difference

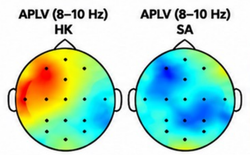
$$\phi_{ij}(t,f) = \arg\left(\frac{R_i(t,f)R_j^*(t,f)}{|R_i(t,f)|\,|R_j(t,f)|}\right)$$



- Phase differences between electrode pairs extracted from normalized RID **coefficients**.

e. Phase-Locking Value (PLV) [3]

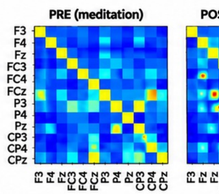
$$PLV_{ij}(f) = \left|\frac{1}{T}\sum_{t=1}^{T} e^{j\phi_{ij}(t,f)}\right| \quad 0 \le PLV_{ij}(f) \le 1$$



- PLV quantifies phase consistency over time within each window.

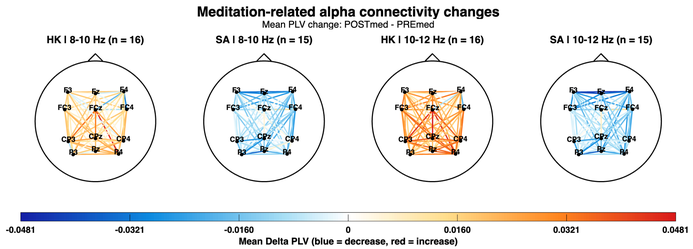
f. Connectivity Matrices & Longitudinal Change

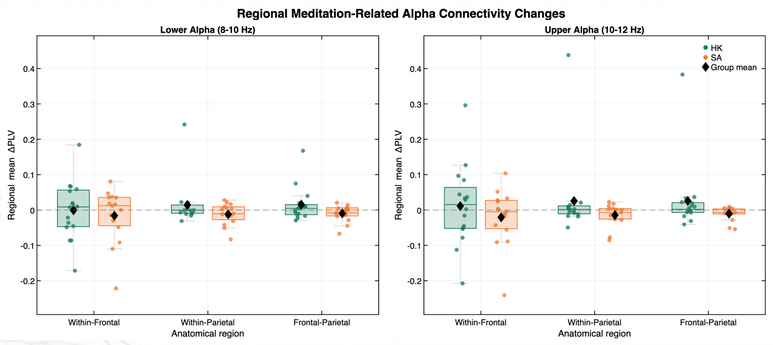
$$\Delta PLV_{ij,med} = PLV_{ij,med}^{POST} - PLV_{ij,med}^{PRE}$$



- **Connectivity Matrices:** Time and Band-Averaged PLV (12 × 12) averaged across subjects.
- **Longitudinal Change:** ΔPLV per participant and electrode pair during meditation.

4A. ALPHA-BAND RESULTS





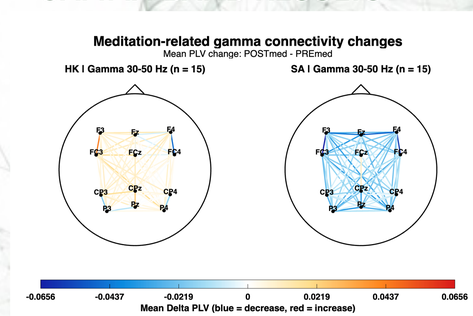
Alpha-Band Connectivity (8–12 Hz)

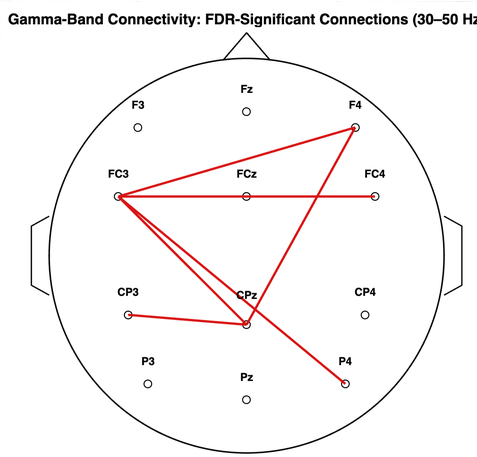
HK showed higher mean ΔPLV than SA in **60/66 lower-α** and **63/66 upper-α** connections.

Regional boxplots showed predominantly positive ΔPLV in HK and negative ΔPLV in SA, with overlapping group distributions.

No edge-wise HK–SA differences survived BH-FDR correction (**all q > 0.05**).

4B. GAMMA-BAND RESULTS





Gamma-Band Connectivity (30–50 Hz)

HK: ΔPLV > 0 in 55/66 connections.

SA: ΔPLV < 0 in 64/66 connections.

6 FDR-significant edges (Wilcoxon rank-sum, q < 0.05), with higher ΔPLV ranks in HK.

5. DISCUSSION

Connectivity Patterns: HK showed increased phase synchronization across alpha and gamma bands, whereas SA exhibited predominantly decreased synchronization.

Frequency-Dependent Effects: Opposing trends were most widespread in upper alpha (10–12 Hz) and gamma (30–50 Hz).

Gamma Band Spatial Organization: Significant connections involved frontocentral and centroparietal regions.

Key Takeaway: Different mantra practices may induce divergent, frequency-specific connectivity changes, highlighting the importance of delineating the mechanism by meditation type.

6. CODE & DATA AVAILABILITY

Scan to access:
- MATLAB analysis code
- EEG dataset
- Supplementary figures

7. REFERENCES

[1] Li, A., et al. (2026). Not all mantra meditations are equal: Emergence of divergent alpha oscillatory dynamics across mantras. bioRxiv. https://doi.org/10.64898/2026.02.26.707862
[2] Li, A., et al. (2026). L-FAME: Longitudinal Focused Attention Meditation EEG Dataset and Benchmark. *arXiv*.
[3] Lachaux, J.-P., et al. (1999). Measuring phase synchrony in brain signals. *Human Brain Mapping, 8*(4), 194–208.
[4] Aviyente, S., Bernat, E. M., Evans, W. S., & Sponheim, S. R. (2011). A phase synchrony measure for quantifying dynamic functional integration in the brain. *Human Brain Mapping, 32*(1), 80–93.